

Marco de Interacción

$$H = \underbrace{\hbar\omega_a \sigma_1^\dagger \sigma_1 + \hbar\omega_a \sigma_2^\dagger \sigma_2}_{H_{\text{libre}}} + \underbrace{\hbar\Omega(\sigma_1^\dagger \sigma_2 + \sigma_1 \sigma_2^\dagger)}_{H_{\text{int}}} + \underbrace{\hbar \varepsilon_0 \cos(\omega_L t) (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2)}_V$$

Queremos pasar a un marco donde la dinámica libre sea eliminada, por ello elegimos al operador unitario de transformación:

$$U = U(t) = e^{-i\omega_L t(\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2)} = e^{-i\omega_L t N}$$

Ec. maestra de Lindblad: $\dot{\rho} = -\frac{i}{\hbar} [H, \rho] + \mathcal{D}\rho$ $\mathcal{D}\rho = \gamma \sum_i (2\sigma_i \rho \sigma_i^\dagger - \sigma_i^\dagger \sigma_i \rho - \rho \sigma_i^\dagger \sigma_i)$

Pasamos al marco de interacción (marco que rota con la frecuencia del láser ω_L):

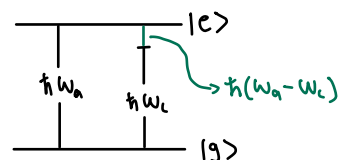
$$\begin{aligned} \rho_I &= U^\dagger \rho U \Rightarrow \dot{\rho}_I = \dot{U}^\dagger \rho U + U^\dagger \dot{\rho} U + U^\dagger \rho \dot{U} = i\omega_L N U^\dagger \rho U + U^\dagger \dot{\rho} U - i\omega_L U^\dagger \rho U N \\ &= i\omega_L [N, \rho_I] + U^\dagger \dot{\rho} U \end{aligned}$$

Veamos qué es $U^\dagger \dot{\rho} U$:

$$\begin{aligned} U^\dagger \dot{\rho} U &= U^\dagger \left(-\frac{i}{\hbar} [H, \rho] + \mathcal{D}\rho \right) U, \quad \mathcal{D}\rho = \gamma \sum_i (2\sigma_i \rho \sigma_i^\dagger - \sigma_i^\dagger \sigma_i \rho - \rho \sigma_i^\dagger \sigma_i) \\ &= -\frac{i}{\hbar} U^\dagger [H, \rho] U + U^\dagger \mathcal{D}\rho U \\ &= -\frac{i}{\hbar} [U^\dagger H U, \rho_I] + U^\dagger \mathcal{D}\rho U \\ &= -\frac{i}{\hbar} [U^\dagger (\hbar\omega_a N + H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U \\ &= -\frac{i}{\hbar} [\hbar\omega_a N, \rho_I] - \frac{i}{\hbar} [U^\dagger (H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U \\ &= -i\omega_a [N, \rho_I] - \frac{i}{\hbar} [U^\dagger (H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U \end{aligned}$$

$$\begin{aligned} \Rightarrow \dot{\rho}_I &= i\omega_L [N, \rho_I] - i\omega_a [N, \rho_I] - \frac{i}{\hbar} [U^\dagger (H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U \\ &= -i(\omega_a - \omega_L) [N, \rho_I] - \frac{i}{\hbar} [U^\dagger (H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U \end{aligned}$$

Sea $\omega = (\omega_a - \omega_L)$ la desintonización del láser



$$\Rightarrow \dot{\rho}_I = -\frac{i}{\hbar} [\hbar\omega N, \rho_I] - \frac{i}{\hbar} [U^\dagger (H_{\text{int}} + V) U, \rho_I] + U^\dagger \mathcal{D}\rho U$$

Ahora veamos qué es $U^\dagger H_{\text{int}} U$:

$$\begin{aligned} [N, \sigma_1^\dagger \sigma_2] &= (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) \sigma_1^\dagger \sigma_2 - \sigma_1^\dagger \sigma_2 (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) \\ &= \sigma_1^\dagger \sigma_1 \sigma_1^\dagger \sigma_2 + \cancel{\sigma_2^\dagger \sigma_2 \sigma_2 \sigma_1^\dagger} - \cancel{\sigma_1^\dagger \sigma_1^\dagger \sigma_1 \sigma_2} - \sigma_2 \sigma_2^\dagger \sigma_2 \sigma_1^\dagger \\ &= |e\rangle\langle e| \langle e| \sigma_1^\dagger |g\rangle\langle g|_2 |g\rangle\langle g|_2 - |g\rangle\langle g|_1 |g\rangle\langle g|_2 |e\rangle\langle e|_1 = |e\rangle\langle g|_1 |g\rangle\langle e|_2 - |e\rangle\langle g|_1 |g\rangle\langle e|_2 = 0 \end{aligned}$$

N y $\sigma_1^\dagger \sigma_2$ conmutan porque N representa el número total de excitaciones y $\sigma_1^\dagger \sigma_2$ crea una en un TLS y la destruye en el otro manteniendo el número total. lo mismo ocurre con $\sigma_1 \sigma_2^\dagger$.

$$\Rightarrow U^\dagger H_{\text{int}} U = H_{\text{int}}$$

Y ahora qué es $U^\dagger V U = e^{i\omega_L t N} \hbar \epsilon (\sigma_1 + \sigma_1^\dagger + \sigma_2 + \sigma_2^\dagger) e^{-i\omega_L t N}$

$$[N, \sigma_1] = (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) \sigma_1 - \sigma_1 (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) = \cancel{\sigma_2^\dagger \sigma_2 \sigma_1} - \sigma_1 \cancel{\sigma_2^\dagger \sigma_2 \sigma_1} - \sigma_1 \sigma_1^\dagger \sigma_1 + \sigma_1^\dagger \sigma_1 \sigma_1$$

$$= -|\uparrow \downarrow \uparrow \rangle \langle \uparrow \downarrow \uparrow| = -|\uparrow \downarrow \rangle \langle \uparrow \downarrow| = -\sigma_1$$

$$\Rightarrow N \sigma_1 - \sigma_1 N = -\sigma_1 \Rightarrow N \sigma_1 = \sigma_1 N - \sigma_1 = \sigma_1 (N-1)$$

$$\Rightarrow U^\dagger \sigma_1 = e^{i\omega_L t N} \sigma_1 = \sigma_1 e^{i\omega_L t (N-1)}$$

$$\Rightarrow U^\dagger \sigma_1 U = \sigma_1 e^{i\omega_L t (N-1)} e^{-i\omega_L t N} = \sigma_1 e^{-i\omega_L t}$$

Similarmente $U^\dagger \sigma_1^\dagger U = \sigma_1^\dagger e^{i\omega_L t}$

$$\Rightarrow U^\dagger (\sigma_1 + \sigma_1^\dagger) U = \sigma_1 e^{-i\omega_L t} + \sigma_1^\dagger e^{i\omega_L t} \quad \wedge \quad U^\dagger (\sigma_2 + \sigma_2^\dagger) U = \sigma_2 e^{-i\omega_L t} + \sigma_2^\dagger e^{i\omega_L t}$$

Por lo tanto:

$$U^\dagger V U = \hbar \epsilon_0 \cos(\omega_L t) (\sigma_1 e^{-i\omega_L t} + \sigma_1^\dagger e^{i\omega_L t} + \sigma_2 e^{-i\omega_L t} + \sigma_2^\dagger e^{i\omega_L t})$$

Recordemos que V es la interacción entre los TLS y un campo externo clásico como el de un láser. La interacción entre un dipolo atómico y un campo eléctrico oscilante es:

$$V = -\hat{d} \cdot \vec{E}(t) \rightarrow E_0 \cos(\omega_L t) \hat{e}, \text{ campo eléctrico clásico}$$

$$\hookrightarrow -q_e \hat{r}_e, \text{ operador dipolar}$$

Para un TLS $\hat{d} = d_{ge} (\sigma + \sigma^\dagger) \hat{u}$ y entonces:

$$V = -d_{ge} E_0 \cos(\omega_L t) (\sigma + \sigma^\dagger)$$

$$\text{sea } \hbar \epsilon_0 = -d_{ge} E_0 \Rightarrow V = \hbar \epsilon_0 \cos(\omega_L t) (\sigma_1 + \sigma_1^\dagger + \sigma_2 + \sigma_2^\dagger)$$

$$\hat{d}_i = d_i (\sigma_i + \sigma_i^\dagger) \hat{u}_i, \quad d_1 = d_2$$

$$V = -d_1 (\sigma_1 + \sigma_1^\dagger) E_0 \cos(\omega_L t) \hat{u}_1 \cdot \hat{e}$$

$$-d_2 (\sigma_2 + \sigma_2^\dagger) E_0 \cos(\omega_L t) \hat{u}_2 \cdot \hat{e}$$

$$= -d E_0 \cos(\omega_L t) [(\sigma_1 + \sigma_1^\dagger) \hat{u}_1 \cdot \hat{e} + (\sigma_2 + \sigma_2^\dagger) \hat{u}_2 \cdot \hat{e}]$$

$$\Rightarrow U^\dagger V U = \hbar \epsilon_0 \cos(\omega_L t) (\sigma_1 e^{-i\omega_L t} + \sigma_1^\dagger e^{i\omega_L t} + \sigma_2 e^{-i\omega_L t} + \sigma_2^\dagger e^{i\omega_L t}), \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$= \hbar \frac{\epsilon_0}{2} (e^{-i\omega_L t} + e^{i\omega_L t}) (\sigma_1 e^{-i\omega_L t} + \sigma_1^\dagger e^{i\omega_L t} + \sigma_2 e^{-i\omega_L t} + \sigma_2^\dagger e^{i\omega_L t})$$

$$= \hbar \epsilon \left(\sigma_1 e^{-2i\omega_L t} + \sigma_1^\dagger + \sigma_2 e^{-2i\omega_L t} + \sigma_2^\dagger + \sigma_1 + \sigma_1^\dagger e^{2i\omega_L t} + \sigma_2 + \sigma_2^\dagger e^{2i\omega_L t} \right)$$

RWA $\rightarrow$ términos que oscilan rápido son despreciados

$$\Rightarrow U^\dagger V U \approx \hbar \varepsilon (\sigma_1 + \sigma_1^\dagger + \sigma_2 + \sigma_2^\dagger)$$

$$\Rightarrow U^\dagger H U = \hbar \omega_a \sigma_1^\dagger \sigma_1 + \hbar \omega_a \sigma_2^\dagger \sigma_2 + \hbar \Omega (\sigma_1^\dagger \sigma_2 + \sigma_1 \sigma_2^\dagger) + \hbar \varepsilon (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2)$$

$$H_I = U^\dagger H U - i \hbar U^\dagger \dot{U}$$

$$= \hbar \omega_a \sigma_1^\dagger \sigma_1 + \hbar \omega_a \sigma_2^\dagger \sigma_2 + \hbar \Omega (\sigma_1^\dagger \sigma_2 + \sigma_1 \sigma_2^\dagger) + \hbar \varepsilon (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2) - \hbar \omega_L N$$

$$= \hbar \omega (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) + \hbar \Omega (\sigma_1^\dagger \sigma_2 + \sigma_1 \sigma_2^\dagger) + \hbar \varepsilon (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2)$$

Con lo que llegamos a:

$$\dot{g}_I = -\frac{i}{\hbar} [\hbar \omega N, g_I] - \frac{i}{\hbar} [H_{int} + \hbar \varepsilon (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2), g_I] + U^\dagger D g U$$

$$= -\frac{i}{\hbar} [H_I, g_I] + U^\dagger D g U$$

Veamos la transformación del disipador:

$$U^\dagger D g U = \mathcal{D} \sum_i U^\dagger (2 \sigma_i g \sigma_i^\dagger - \sigma_i^\dagger \sigma_i g - g \sigma_i^\dagger \sigma_i) U$$

$$U^\dagger \sigma_i g \sigma_i^\dagger U = U^\dagger \sigma_i U g_I U^\dagger \sigma_i^\dagger U = \sigma_i e^{-i \omega_L t} g_I \sigma_i^\dagger e^{i \omega_L t} = \sigma_i g_I \sigma_i^\dagger$$

Para los demás términos se anulan estas fases de manera similar.

$$\Rightarrow U^\dagger D g U = \mathcal{D} \sum_i (2 \sigma_i g_I \sigma_i^\dagger - \sigma_i^\dagger \sigma_i g_I - g_I \sigma_i^\dagger \sigma_i)$$

La ec. maestra de Lindblad en el marco de interacción es:

$$\dot{g}_I = -\frac{i}{\hbar} [H_I, g_I] + D g_I$$

$$\text{donde } H_I = \hbar \omega (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) + \hbar \Omega (\sigma_1^\dagger \sigma_2 + \sigma_1 \sigma_2^\dagger) + \hbar \varepsilon (\sigma_1^\dagger + \sigma_1 + \sigma_2^\dagger + \sigma_2),$$

$$D g_I = \mathcal{D} \sum_i (2 \sigma_i g_I \sigma_i^\dagger - \sigma_i^\dagger \sigma_i g_I - g_I \sigma_i^\dagger \sigma_i)$$

Comentarios sobre dinámicas:

1: $g_s(0) = g_I(0)$, los estados iniciales son iguales en ambos marcos ya que

$$U(0) = e^{-i \omega_L \cdot 0 \cdot N} = \mathbb{1} \Rightarrow g_I(0) = U^\dagger(0) g_s(0) U(0) = \mathbb{1} g_s(0) \mathbb{1} = g_s(0)$$

2: Las poblaciones son iguales en ambos marcos:

$$N |gg\rangle = (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) |gg\rangle = 0$$

$$N |eg\rangle = (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) |eg\rangle = |eg\rangle$$

$$N |ge\rangle = |ge\rangle$$

$$N |ee\rangle = (\sigma_1^\dagger \sigma_1 + \sigma_2^\dagger \sigma_2) |ee\rangle = |ee\rangle + |ee\rangle = 2 |ee\rangle$$

$$\Rightarrow U|gg\rangle = |gg\rangle, U|ge\rangle = e^{-i\omega_c t} |ge\rangle, U|eg\rangle = e^{-i\omega_c t} |eg\rangle, U|ee\rangle = e^{-2i\omega_c t} |ee\rangle$$

Las poblaciones tienen la forma $\langle j | \rho_I(t) | j \rangle = \langle j | U^\dagger(t) \rho_S(t) U(t) | j \rangle$ para cierto estado $|j\rangle = |gg\rangle, |ge\rangle, |eg\rangle, |ee\rangle$

Dado que $U(t)|j\rangle = c_j |j\rangle$ y $U^\dagger(t)|j\rangle = c_j^* |j\rangle$ donde c_j es la fase:

$$\begin{aligned} \langle j | \rho_I(t) | j \rangle &= \langle j | c_j^* \rho_S(t) c_j | j \rangle = |c_j|^2 \langle j | \rho_S(t) | j \rangle \\ &\quad \rightarrow 1 \text{ para todos los casos} \\ &= \langle j | \rho_S(t) | j \rangle \end{aligned}$$

3: El módulo de las coherencias son iguales en ambos marcos:

Las coherencias son los elementos no-diagonales

$$\langle j | \rho_I(t) | k \rangle = \langle j | U^\dagger(t) \rho_S(t) U(t) | k \rangle, \quad j \neq k$$

$$U(t)|j\rangle = e^{-i\omega_c t n_j} |j\rangle, \text{ donde } n_j \text{ es el número de excitaciones del estado } |j\rangle$$

$$\Rightarrow \langle j | \rho_I(t) | k \rangle = e^{i\omega_c t (n_j - n_k)} \langle j | \rho_S(t) | k \rangle$$

$$\text{Tomando el valor absoluto: } |\langle j | \rho_I(t) | k \rangle| = |e^{i\omega_c t (n_j - n_k)} \langle j | \rho_S(t) | k \rangle|$$

$$\Rightarrow |\langle j | \rho_I(t) | k \rangle| = |\langle j | \rho_S(t) | k \rangle|$$

4: La concurrencia es igual en ambos marcos:

La concurrencia es invariante bajo transformaciones unitarias y locales (que afectan separadamente a cada qubit) como lo es $U(t)$:

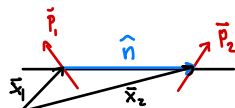
$$U(t) = U_1(t) \otimes U_2(t) \text{ donde } U_i(t) = e^{-i\omega_c t \sigma_i^+ \sigma_i}$$

$$\text{Derivación de } H_{\text{int}} = \hbar \Omega (\sigma_1^+ \sigma_2 + \sigma_1 \sigma_2^+)$$

Recordemos que el operador dipolar para 1 TLS tiene la forma: $\hat{d}_i = d_i (\sigma_i + \sigma_i^+) \hat{e}_i$

Y que la energía potencial clásica entre 2 dipolos es:

$$W_{12} = \frac{\vec{p}_1 \cdot \vec{p}_2 - 3(\hat{n} \cdot \vec{p}_1)(\hat{n} \cdot \vec{p}_2)}{4\pi\epsilon_0 |\vec{x}_1 - \vec{x}_2|^3}$$



En M.C esta energía se vuelve el operador hamiltoniano de interacción:

$$H_{\text{int}} = \frac{1}{4\pi\epsilon_0} \frac{\hat{d}_1 \cdot \hat{d}_2 - 3(\hat{d}_1 \cdot \hat{r})(\hat{d}_2 \cdot \hat{r})}{r^3}, \quad \vec{r}: \text{vector de separación entre los dipolos}$$

Sustituyendo los operadores dipolares:

$$H_{int} = \frac{1}{4\pi\epsilon_0} \frac{[d_1(\sigma_1 + \sigma_1^\dagger)\hat{e}_1] \cdot [d_2(\sigma_2 + \sigma_2^\dagger)\hat{e}_2] - 3\{[d_1(\sigma_1 + \sigma_1^\dagger)\hat{e}_1] \cdot \hat{r}\}\{[d_2(\sigma_2 + \sigma_2^\dagger)\hat{e}_2] \cdot \hat{r}\}}{r^3}$$

$$= \frac{1}{4\pi\epsilon_0 r^3} [\hat{e}_1 \cdot \hat{e}_2 - 3(\hat{e}_1 \cdot \hat{r})(\hat{e}_2 \cdot \hat{r})] d_1 d_2 (\sigma_1 + \sigma_1^\dagger)(\sigma_2^\dagger + \sigma_2)$$

$$(\sigma_1 + \sigma_1^\dagger)(\sigma_2^\dagger + \sigma_2) = \sigma_1 \sigma_2^\dagger + \sigma_1 \sigma_2 + \sigma_1^\dagger \sigma_2^\dagger + \sigma_1^\dagger \sigma_2$$

$\sigma_1 \sigma_2$ y $\sigma_1^\dagger \sigma_2^\dagger$ oscilan rápidamente en la imagen de Heisenberg por lo que usando RWA pueden ser despreciados.

$$\Rightarrow (\sigma_1 + \sigma_1^\dagger)(\sigma_2^\dagger + \sigma_2) \approx \sigma_1 \sigma_2^\dagger + \sigma_1^\dagger \sigma_2$$

$$\text{Sea } \hbar\Omega = \frac{d_1 d_2}{4\pi\epsilon_0 r^3} [\hat{e}_1 \cdot \hat{e}_2 - 3(\hat{e}_1 \cdot \hat{r})(\hat{e}_2 \cdot \hat{r})]$$

Obtenemos la forma de H_{int}

Magnitudes

$$\omega_a \sim 3.84 \times 10^{14} \text{ Hz} \rightarrow \text{línea D2 del Rubidio 87 (luz roja de 780 nm)}$$

$$\omega_L \approx \omega_a \rightarrow \text{Resonancia } \omega_a \gg |\omega|, \omega = \omega_a - \omega_L. \omega \sim \Gamma, \Omega, \mathcal{E} (10^6 \text{ Hz})$$

$$\Gamma_0 \sim 2\pi \times 6 \times 10^6 \text{ Hz} \approx 3.77 \times 10^7 \rightarrow \text{Vida media del } |e\rangle \text{ de Rb 87 } \tau = 26.6 \text{ ns}$$

$$\Omega_0 \sim \text{KHz} \rightarrow \text{Distancia de átomos y dipolo atómico}$$

$$\mathcal{E}_0 \sim 2\pi \times 10^6 \text{ Hz} \rightarrow \text{Intensidad del campo externo dipolo atómico}$$

$$\omega = \omega_a - \omega_L, \frac{\Gamma}{\omega} = 0.2 \Rightarrow \omega = \frac{\Gamma}{0.2} = \omega_a - \omega_L \Rightarrow \omega_L = \omega_a - \frac{\Gamma}{0.2} \approx 3.83999 \times 10^{14}$$

$$\Rightarrow \omega_0 = 1.885 \times 10^8 \text{ Hz}$$

Unidades adimensionales:

$$\omega = \frac{\omega_0}{\omega_0} = 1$$

$$\Gamma = \frac{\Gamma_0}{\omega_0} = 0.2$$

$$\Omega = \frac{\Omega_0}{\omega_0} = 0.4 ??? \Rightarrow \Omega_0 = 1.7 \times 10^8 \text{ Hz}$$

$$\mathcal{E} = \frac{\mathcal{E}_0}{\omega_0} = 0.033$$

$$\Gamma = \frac{\omega_a^2 |d|^2}{3\pi\epsilon_0 \hbar c^3} \Rightarrow d^2 = 3\pi\epsilon_0 \hbar c^3 \frac{\Gamma}{\omega_a^3}$$

$$S_1 \quad \begin{array}{c} \text{d}_1 \rightarrow \text{f} \rightarrow \text{d}_2 \\ \text{red} \quad \text{blue} \quad \text{red} \end{array} \Rightarrow \hbar \Omega = \frac{3\pi\epsilon_0 \hbar c^3}{4\pi\epsilon_0 r^3} \frac{\Gamma}{\omega_a^3} (-2) \Rightarrow \Omega = -\frac{3}{2} \left(\frac{c}{r\omega_a} \right)^3 \Gamma$$

$$\text{Si queremos } \Omega = -1.7 \times 10^8 \text{ Hz} \Rightarrow r = \sqrt[3]{-\frac{3}{2} \left(\frac{c}{\omega_a} \right)^3 \frac{\Gamma}{\Omega}} \approx 5 \times 10^{-7} \text{ m} = 0.5 \mu\text{m}$$

$$\text{Trampas realistas } r = 2 \mu\text{m} \Rightarrow \Omega \approx -3 \times 10^6 \text{ Hz} = \text{MHz} \Rightarrow \Omega \rightarrow \frac{\Omega}{\omega_a} = -0.1$$

Concurrencia:

$$R = \tilde{S} \tilde{S}^\dagger, \tilde{S} = S_Y \tilde{S}^\dagger S_Y, S_Y = \sigma_Y \otimes \sigma_Y$$

$$C(S) = \max(0, \sqrt{\lambda_{R_1}} - \sqrt{\lambda_{R_2}} - \sqrt{\lambda_{R_3}} - \sqrt{\lambda_{R_4}}), \lambda_{R_i} \text{ eigenvalores de } R \text{ de mayor a menor}$$

$$R_I = U^\dagger R U = U^\dagger \tilde{S} \tilde{S}^\dagger U = \tilde{S}_I U^\dagger S_Y \tilde{S}^\dagger S_Y U = \tilde{S}_I U^\dagger S_Y U^\dagger U \tilde{S}^\dagger U^\dagger U S_Y U$$

$$U \tilde{S}^\dagger U^\dagger = U^\dagger \tilde{S}^\dagger U^{\dagger\dagger} = (U^\dagger \tilde{S} U)^T = \tilde{S}_I^T, U \text{ y } U^\dagger \text{ son diagonales}$$

$$U^\dagger S_Y U^\dagger = U^\dagger \sigma_Y \otimes \sigma_Y U^\dagger = (U_1^\dagger \otimes U_2^\dagger) (\sigma_Y \otimes \sigma_Y) (U_1^\dagger \otimes U_2^\dagger)$$

$$= U_1^\dagger \sigma_Y U_1^\dagger \otimes U_2^\dagger \sigma_Y U_2^\dagger = e^{i\omega_c t} \sigma_Y \otimes \sigma_Y e^{i\omega_c t} = e^{2i\omega_c t} \sigma_Y \otimes \sigma_Y$$

$$= \tilde{S}_I S_Y \tilde{S}_I^T S_Y$$

$$\searrow e^{i\omega_c t} \sigma_Y$$

$$N(\mathcal{S}_\theta(t)) = \frac{1}{4} e^{-4t\gamma} (3 - ze^{2\gamma t} + |1 - ze^{i\gamma t}|) = \frac{1}{2} e^{-4\gamma t}$$

$$N(\mathcal{S}_\psi(t)) = \frac{1}{2} e^{-2\gamma t} (1 - e^{2\gamma t} + \sqrt{2 - ze^{2\gamma t} + e^{4\gamma t}}) =$$